

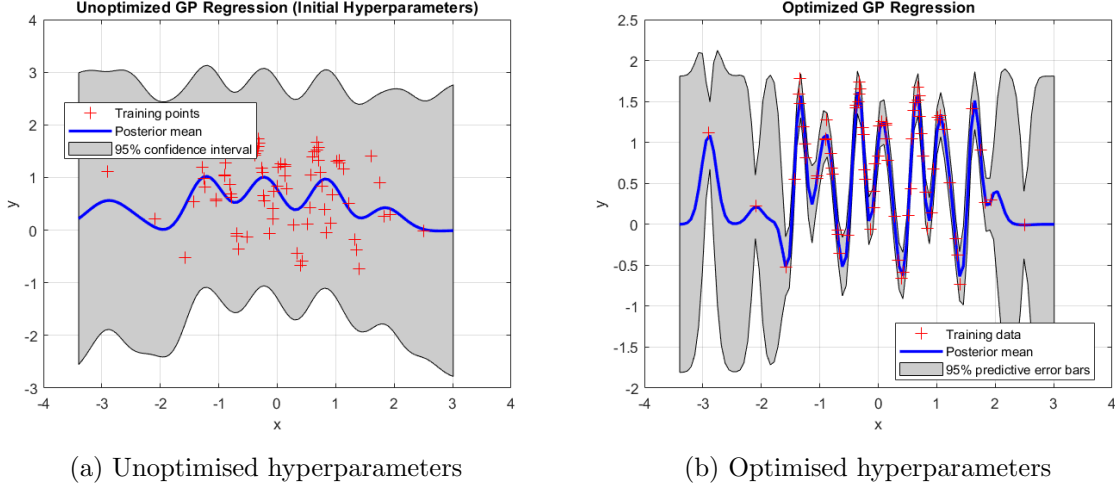


Figure 1: Posterior mean and predictive error bars

- (a) Figure 1 highlights the effect of training a Gaussian Process using the Squared Exponential (SE) kernel, defined in Equation 1 below, where the output variance v^2 behaves as a scale factor to describe the average distance of your function away from its mean.

$$k_{SE}(x, x') = v^2 \exp\left(-\frac{(x - x')^2}{2l^2}\right) \quad (1)$$

We minimize the negative log marginal likelihood to obtain the parameters of Table 1.

	Length scale (l)	Signal std (v)	Noise std (σ_n)
Initial	0.3679	1	1
Optimised	0.1282	0.897	0.1178

Table 1: Initial and optimised hyperparameters with SE covariance function

From Table 1, we observe a clear reduction in length scale l in the optimised case. For reduced length scales, we can use Equation 1 to infer that the covariance between data points decreases rapidly as move away from each other, allowing the model to follow fast-changing patterns, as we can see on Figure 1b. The increased flexibility at smaller length scales increases the quality of the fit, resulting in lower optimal values of σ_n (Table 1). On the other hand, a larger length scale in the Gaussian kernel creates stronger correlations between distant points. This means function values remain dependent on each other even at greater distances along the x-axis. As a result, the blue curve fails to adapt to rapid variations in the data, leading to a poor fit on Figure 1a.

We can also observe that the 95% predictive uncertainty bands are narrowest at observed data points and widen in regions without observations. The small inferred length scale causes uncertainty to increase sharply with distance from observations, creating distinct valleys in the error bands at $x = \pm 2.5$ on Figure 1b. The length scale is meant to represent how far you need to move in the input space for the function values to become uncorrelated [3]: in general, it is hard to extrapolate more than l units away from your data [3]. This reflects how the covariance function tightly constrains predictions near known data points.

- (b) To explore whether the hyperparameter optimum is unique, we can perform multiple optimizations from different starting points in the hyperparameter space, and observe

whether or not we find different optimas. Using grid search, we were able to find a non-exhaustive 3 distinct local optima with considerably different length scales (see Table 2). Note that for all points, $\text{hyp.lik} = 0.0$.

Optimum number	Example hyp.cov	l	v	σ_n	$\log p(y x)$
1	[-1.0, 0.0]	0.128	0.897	0.118	-11.899
2	[1.0, 1.0]	8.04	0.696	0.663	-78.220
3	[10.0, 0.0]	2.203×10^4	0.744	0.667	-78.337

Table 2: Different optima found through grid search with example starting points.

The log marginal likelihood, given by Equation 2 [3] is used to quantify the quality of the fit.

$$\log p(\mathbf{y}|\mathbf{x}) = -\frac{1}{2}\mathbf{y}^T(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}\mathbf{y} - \frac{1}{2}\log |\mathbf{K} + \sigma_n^2\mathbf{I}| - \frac{n}{2}\log(2\pi) \quad (2)$$

From Table 2, we observe the log marginal likelihood of optimum 1 is significantly larger (over $e^{66} \approx 10^{29}$) than the 2 others. Optimum 1 seems to be the global optimum of our log marginal likelihood, whereas optima 2 & 3 are only local optima. Consequently, we can confidently claim that the model with the shorter length scale is the best fit.

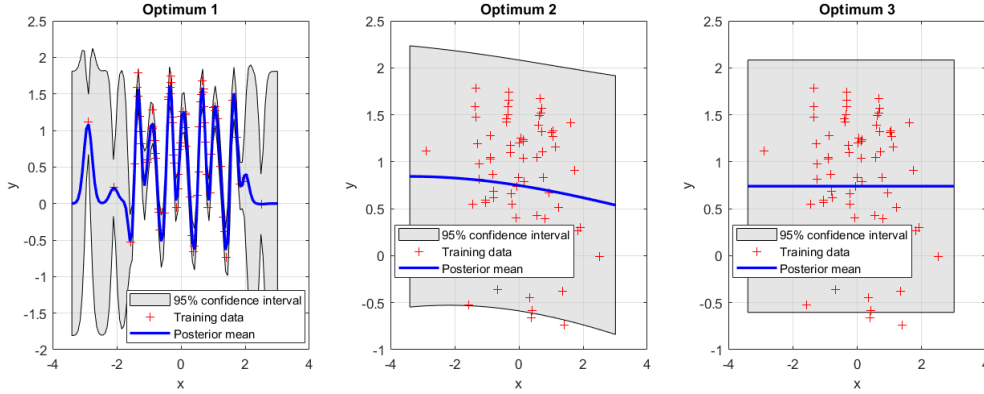


Figure 2: Posterior mean and predictive error bars with various hyperparameters

On Figure 2, Optimum 1 corresponds to our solution from a), capturing rapid variations with a short length scale. Optimum 2 represents a smoother function with a larger length scale, while Optimum 3 approaches the limit $l \rightarrow \infty$, essentially modeling the data as pure noise around a constant, hence the larger σ_n . Graphically, optimum 1 seems to provide a much better fit than the 2 other optima, which reinforces the argument developed in the previous paragraph.

- (c) The periodic kernel, derived by David MacKay, is defined similarly as the covariance function in Equation 1, but with an additional hyperparameter p , the period, which represents distances between repetitions of the function [1].

$$k_{Per}(x, x') = v^2 \exp\left(-\frac{2}{l^2} \sin^2\left(\frac{\pi|x - x'|}{p}\right)\right) \quad (3)$$

The kernel derives its name well: we can observe the periodic pattern of our posterior mean on Figure 3, which seems to have periodicity p . Whereas we noticed large error bars when further away from data in Figure 1b, this phenomenon is not present in Figure 3. Since the covariance doesn't just decay with increasing distance from data $|x - x'|$,

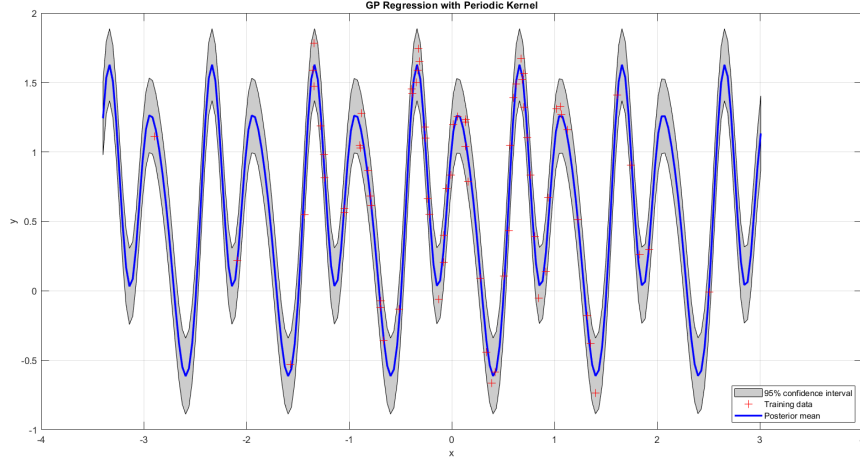


Figure 3: Posterior mean and predictive error bars with periodic covariance function

the periodicity helps make confident predictions on the posterior mean’s behaviour, even when data is missing. For example, we are able to use the function’s behaviour around $x = -1$, $x = 0$, and $x = 1$ to confidently predict the behaviour around $x = 2$.

Hyperparameters	Period	Length-scale	Signal std	Noise std	Log marginal likelihood
Optimised	1.0444	0.9989	1.2359	0.1094	35.2508

Table 3: Optimized hyperparameters for the model.

Just like in question a), the posterior mean’s good fit to the data leads to a small σ_n (Table 3). Despite the accurate fit, we can’t help but notice how much higher the log marginal likelihood is for our periodic model (35 compared to -12 on logarithmic scales, ie), compared to fitting with the Radial Basis Function kernel. Although the Squared Exponential kernel is universal and can be integrated against most functions [1], the periodic kernel only works on periodic functions. The fact the fit is so accurate supports our evidence that the data-generating mechanism is periodic. However, we are not able to pronounce ourselves on the veracity of this statement outside of the domain $[-3.5; 3]$.

- (d) Our composite covariance function is called Locally Periodic covariance function [1]. It has the form found in Equation 4, where $k_{SE}(x, x')$ and $k_{Per}(x, x')$ are defined in Equations 1 and 3 respectively.

$$k_{LocalPer}(x, x') = k_{SE}(x, x') * k_{Per}(x, x') \quad (4)$$

Cholesky decomposition can only occur if the covariance matrix is positive definite. Mathematically, adding a small diagonal matrix makes sure that the eigenvalues are larger than $10^{-6} > 0$.

We first plot 3 random noise-free sample functions from a GP with composite covariance function in Figure 4. Although not strictly periodic¹ like in the previous question, we clearly notice a periodic behaviour, only with changing amplitudes. This adds flexibility to our model compared to the periodic one [1].

Table 4 below transcribes the important hyperparameter values. Having such small length scale l_{Per} leads to significant variance of the periodic component but smooth amplitude

¹A function f is strictly periodic if $\exists p \in \mathbb{R} \setminus \{0\} \forall x \in \mathbb{R}, f(x) = f(x + p)$

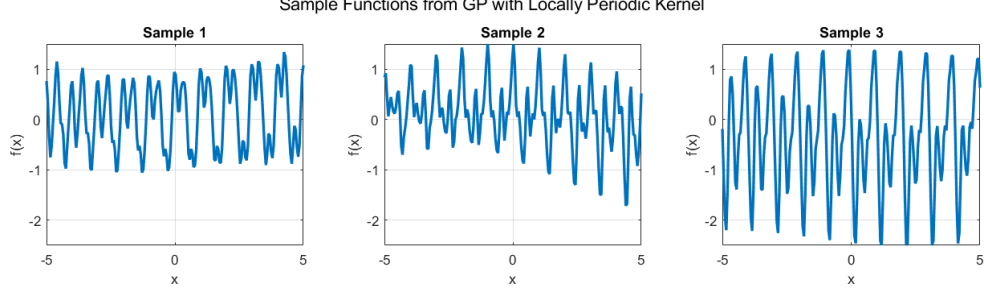


Figure 4: 3 sample functions from a GP with composite covariance function

changes due to a much larger l_{SE} length scale.

l_{Per}	v_{Per}	p	l_{SE}	v_{SE}
0.607	1	1	7.389	1

Table 4: Covariance Hyperparameters for Locally Periodic Kernel

Figure 5 further illustrates the similarities and differences of Periodic and Locally Periodic kernels. Both matrices keep the periodic striped pattern, but the locally periodic covariance matrix fades off as we move further away from a given point (top right and bottom left of Figure 5).

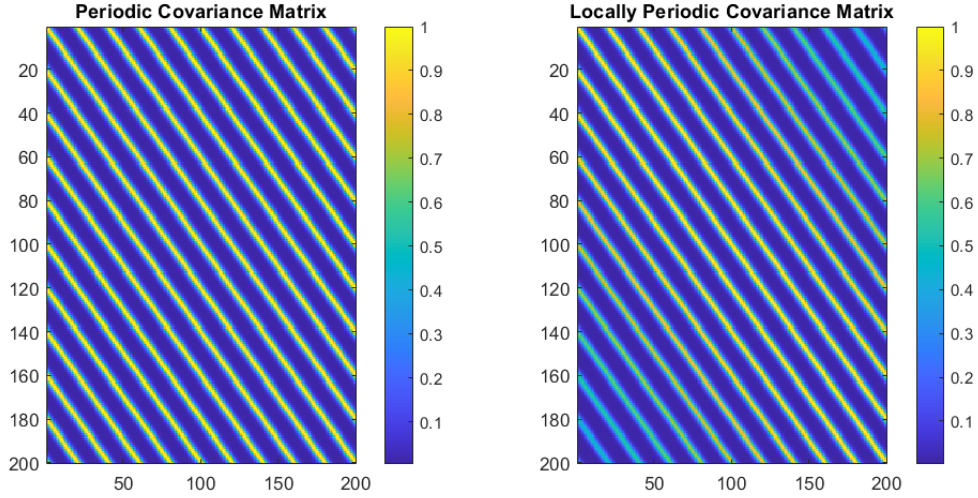


Figure 5: Covariance matrix heatmap for Periodic and Locally Periodic kernels

- (e) Given we now have several inputs, we could use automatic relevance determination (ARD) covariance functions for feature/variable selection, as it effectively removes irrelevant inputs [3]. These SE ARD kernels have the form of Equation 5 below [2]:

$$k_{ARD}(\mathbf{x}, \mathbf{x}') = v^2 \exp \left(- \sum_{d=1}^D \frac{(x_d - x'_d)^2}{2l_d^2} \right) \quad (5)$$

The log marginal likelihood of a Gaussian was already defined in Equation 2. The first term is the data fit term: this value decreases as the model fits the data better. The second term of the equation represents a complexity penalty, meant to encourage the use

of simpler models. The benefits of this penalty term are to avoid potential overfitting problems but also to increase the interpretability and explainability of the model.

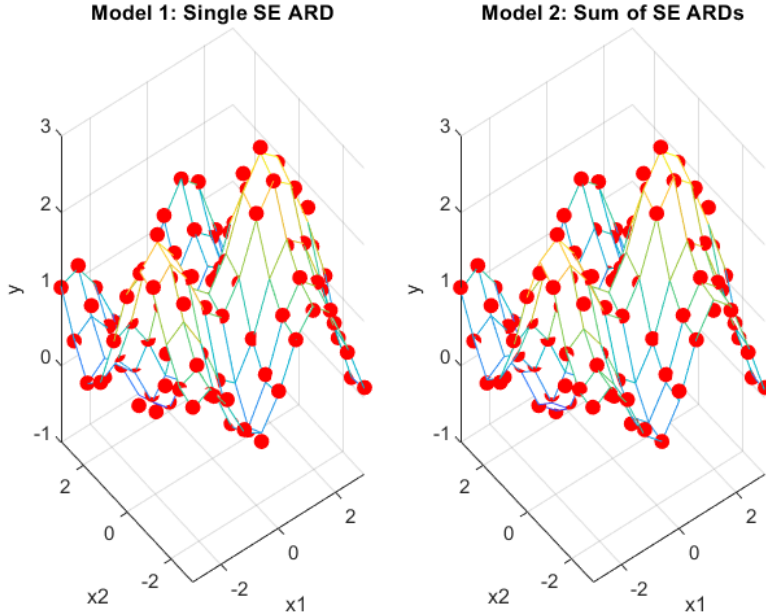


Figure 6: Trained Gaussian Processes with single SE ARD vs sum of SE ARDs covariance functions, with data points

From Figure 6, we notice how similar the data fit for both models looks graphically, leading to similar data fit values in Table 5. The single model is less complex as there are fewer hyperparameters. However, the complexity values in Table 5 are greater for the single SE ARD compared to the sum of SE ARDs. Using Occam’s Razor, we will prioritise the less complex model. The sum of SE ARDs model is best, which is also illustrated by the larger log marginal likelihood value in Table 5.

Model	Data fit	Complexity	l_1	l_2	Signal var.	σ_n	lml
Single SE-ARD	-60.50	-190.91	1.51	1.29	1.23	0.103	19.22
Sum of SE-ARDs	-60.53	-238.12	(1043.79, 0.98)	(1.45, 1719.84)	0.50, 1.23	0.098	66.35

Table 5: Optimized hyperparameters for both single & sum of SE ARD as well as decomposition of the log marginal likelihood into data fit and complexity terms

References

- [1] D. K. Duvenaud. “Automatic Model Construction with Gaussian Processes”. PhD thesis. Cambridge, UK: University of Cambridge, June 2014, pp. 8–12.
- [2] C. E Rasmussen. *4F13 Gaussian Process lecture notes*. Oct. 2024.
- [3] C. E. Rasmussen & C. K. I. Williams. *Gaussian Processes for Machine Learning*. Cambridge, Massachusetts: MIT Press, 2006, pp. 19, 106. URL: <http://www.GaussianProcess.org/gpml>.